Homework week 5 & 6

Practical problems

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Until next week (20.5)

Exercise

Excericse 1

Use the `trees` data set (located in `data/trees.rds`) and select only spruce, pine and oak trees for the year 2020. First, investigate graphically, if you think that *bio18* (percipitation in the warmest month) could have a linear effect on `mean_loss`. Then fit a linear model for this relationship and give an interpretation of the estimated intercept, slope and variance parameter. Fit two further models that 1) include species and an interaction of species and `bio18`. Again look at the estimated coefficients, what do they mean?

Reading

- Heinze & Dunkler 2016: Five myths about variable selection, *Transplant International*: https://onlinelibrary.wiley.com/doi/10.1111/tri.12895

Until week 9 (10.6.2026)

Exercise

Exercise 2

Visualize the three models from Exericise 1 (give each species its own color) and show the values for `bio18` on the x-axis and the predicted mean loss on the y-axis. Can you create three variants of the plot? Once without uncertainty, once withe a prediction interval and once with a confidence interval?

Finally, compare the three models with the adjusted R^2, which one do you think is better?

Exercise 3 (optional)

Try to estimate the coefficients of the model (`mean_damage ~ bio18 * species`) from exericse 1 in four different ways:

1. With ordinary least square using the `lm()`-function.
2. Using matrix algebra.
3. With the `brms`-package using the function `brm()`.
4. With Stan (optionally, also with matrix notation).

Reading (optional)

One of my favorite papers on the topic ist:

- Tredennick et al. 2021: A practical guide to selecting models for exploration, inference, and prediction in ecology, *Ecology*: https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecy.3336